Type MS · Distance travelled by integrating velocity

Answer 1 of 8 · Question 3 part b · 4 marks

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3(a)	$\frac{dv}{dt} = 2t - 4$ Accel $= 2\ \left(m/s^2\right)$	M1 A1 (2)
(b)	$s = \int_0^6 \left(t^2 - 4t + 7\right)dt = \left[\frac{t^3}{3} - 2t^2 + 7t\right]_0^6$ $= \frac{6^3}{3} - 2\times 6^2 + 7\times 6 = 42\ (m)$	M1A1 dM1 A1cao (4) [6]

(a)	
M1	Attempt to differentiate the expression for v. Power of t to decrease in at least one term and increase in none.
A1	Substitute $t = 3$ and obtain correct acceleration – units may be missing
(b)	
M1	Attempt to integrate the expression for v. Power of t to increase in at least one term and decrease in none. Ignore limits if shown. Constant not needed for indefinite integration.
A1	Correct integration. Limits/constant not needed.
dM1	**Either** substitute the limits 0 and 6 **or** use $s = 0$, $t = 0$ to obtain a value for the constant and substitute $t = 6$ in the complete expression. (Substitution of 0 can be implied if the result would have been 0) Depends on the previous M mark If more values of t are substituted **and results used** award M0
A1cao	$S = 42$ (m)
NB	Ans 42 w/o working scores 4/4 (Done on a calculator)

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Answer 1 of 8 · Question 3 part b · 4 marks

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Question Number	Scheme	Marks
5	$(x+2y=17)\quad x=\frac{36}{y}\quad \left(\text{or } y=\frac{36}{x}\right)$	M1
	$\frac{36}{y}+2y=17,\quad 36+2y^2=17y\quad (\text{or } 72+x^2=17x)$	M1
	$2y^2-17y+36\ (=0)\qquad (\text{or } x^2-17x+72=0)$	A1
	$(y-4)(2y-9)=0\qquad (\text{or } (x-8)(x-9))$	
	$y=4\quad x=9$	dM1A1
	$y=4\frac{1}{2}\quad x=8$	A1 (6)
M1 M1 A1 M1 A1 A1	Rearrange $xy = 36$ to $x = ...$ or $y = ...$ Eliminate x or y from the linear equation and obtain a 3TQ, $=0$ not needed Correct 3TQ, terms in any order. $=0$ not needed Solve their 3TQ by any valid method. Obtain at least one value for y or x **Either** 2 correct values for x or y **or** a correct (x, y) pair Both pairs correct and pairing clear.	
ALT:	The following method may possibly be seen: $xy + x + 2y = 53 \Rightarrow 36 + x + 2y = 53 \Rightarrow x + 2y = 17$ and $xy = 36$ or $x \times 2y = 72$	M1
	Hence x and $2y$ are the roots of the equation $z^2 - 17z + 72 = 0$	M1A1
	$(z - 9)(z - 8) = 0 \Rightarrow z = 9$ or 8	M1
	So $x = 8$ $y = 4.5$ or $x = 9$ $y = 4$	A1A1 [6]
M1 M1 A1 M1 A1 A1	Substitute $xy = 36$ in the linear equation to obtain $x + 2y = 17$ and $xy = 36$ oe Obtain a 3TQ with roots x and $2y$ Correct 3TQ Solve their 3TQ by any valid method. Obtain at least one value for *for the roots* **Either** 2 correct values for x or y **or** a correct (x, y) pair Both pairs correct and pairing clear.	
	Special Case $x+2y=17\quad xy=36$ Use $xy=36$ in the other equation to obtain $x+2y=17$ $\Rightarrow x=9\quad y=4$ By inspection: Score M1M0A0M1A1A0 (Must see $x+2y=17$; otherwise no marks) If the second answer is also obtained correctly by inspection, award all marks	

Type MS · Distance travelled by integrating velocity

Answer 1 of 8 · Question 3 part b · 4 marks

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Question Number	Scheme	Marks
6(a)(i)	$\frac{dy}{dx} = 4e^{2x} + 2(4x-3)e^{2x}$	M1A1A1 (3)
(ii)	$(4x-3)\frac{dy}{dx} = (4x-3)(8x-2)e^{2x} = (8x-2)y$ *	M1A1cso (2)
(b)	$\frac{dy}{dx} = \frac{5\cos 5x \times (x-3)^2 - \sin 5x \times 2(x-3)}{(x-3)^4}$	M1A1A1 (3)
ALT	Using product rule:	
	$y = (x-3)^{-2}\sin 5x$	M1
	$\frac{dy}{dx} = -2(x-3)^{-3}\sin 5x + 5(x-3)^{-2}\cos 5x$	A1A1
		[8]

(a)(i)

M1 Use product rule to differentiate the given expression. Must have 2 terms added. One to be of the form ke^{2x} and the other of the form $k'(4x-3)e^{2x}$ where k' = 1 or 2

A1 Either term correct

A1 Second term correct

NB No simplification needed for these 3 marks

ALT $y = 4xe^{2x} - 3e^{2x} \Rightarrow \frac{dy}{dx} = 4e^{2x} + 8xe^{2x} - 6e^{2x}$

M1 Expand the given expression and differentiate using the product rule for $4xe^{2x}$

A1 Any 2 terms correct; A1 Third term correct.

NB No simplification needed for these 3 marks

(ii)

M1 Use their result from (i) to obtain an expression for $(4x-3)\frac{dy}{dx}$. No need to simplify.

A1cso Correct **given** result obtained with no errors in the working.
Can start with LHS and show equal to the RHS or vice versa or can start with each side and "meet in the middle"

(b)

M1 Attempt the quotient rule. The denominator must be $(x-3)^4$ and the numerator must be of the form $\left(k\cos 5x \times (x-3)^2 - \sin 5x \times l(x-3)\right)$ $k = \pm 5$ or ± 1, $l = 1$ or 2
(ie sine may have been differentiated to − cosine)

A1 One fully correct term in numerator.

A1 All fully correct.

ALT

M1 Rewrite without a quotient and apply the product rule obtaining 2 terms of the form shown

A1 Either term correct

A1 Second term correct
No need to simplify

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Answer 1 of 8 · Question 3 part b · 4 marks

2019 JUN · Paper 02 · extract 4 of 11 · find part b in this mark scheme extract

Question Number	Scheme	Marks
7(a)	(i) $a = 9$	B1
	(ii) $d = 4$	B1 (2)
(b)	(i) $a = 4$	B1
	(ii) $r = 3$	B1 (2)
(c)	$A_{14} = \frac{14}{2}(2\times 9 + 13\times 4)$ or $\frac{14}{2}(9+61)$, $= 490$	M1, A1
	$"490" - 6 = \frac{4(3^n - 1)}{3-1}$	M1
	$3^n = 243 \quad n = 5$	ddM1A1 (5) [9]

(a)
B1 Correct value, no working or explanation needed
B1 Correct value, no working or explanation needed
(b)
B1 Correct value, no working or explanation needed
B1 Correct value, no working or explanation needed
(c)
M1 Use either formula for the sum of an arithmetic series with their a and d (if needed) and obtain a value for the sum of the first 14 terms
A1 Correct value for the sum
M1 Subtract 6 from their sum (explicitly or implicitly) and equate to the sum of the first n terms of the geometric series obtained using their a and r
ddM1 Solve their equation by a correct method. No method need be shown but must reach n = ... Depends on both M marks above
A1 Correct value for n obtained

ALT For the last 3 marks:
M1 Subtract 6 from their sum and generate at least the first 5 terms of the geometric series.
ddM1 Sum their terms until at least "484" is reached
A1 Correct answer (5) obtained from correct work.

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Answer 1 of 8 · Question 3 part b · 4 marks

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Question Number	Scheme	Marks
8(a)	$AB^2 = 4^2 + 2^2,\ BC^2 = 2^2 + 6^2,\ AC^2 = 2^2 + 4^2$	M1 (any one)
	(i) $AB = \sqrt{20}$ (4.47) (ii) $BC = \sqrt{40}$ (6.32) (iii) $AC = \sqrt{20}$ (4.47) or equivalents	A1A1A1 (4)
(b)	Any complete method for finding one of the angles: eg $AB^2 + AC^2 = BC^2 \Rightarrow \angle A = 90°$ or use trigonometry	M1
	$\angle A = 90°,\ \angle B = \angle C = 45°$	A1, A1 (3)
I	(centre at midpoint of BC) $(5, 5)$	M1A1 (2)
(d)	Radius $= \frac{1}{2}BC = \frac{1}{2}\sqrt{40} = \sqrt{10}$ (Working for (d) may be seen in a previous part)	M1A1 (2) [11]

(a)	
M1	Use Pythagoras with a plus sign to obtain AC^2, BC^2 or AC^2. If the answer is incorrect it must be clear that the correct coordinates have been used correctly.
A1A1A1	Award A1 for each correct length. Ignore labels (i), (ii) and (iii). Award M1A1A1A1 / M1A1A1A0 / M1A1A0A0 as appropriate.
SC:	If there is no working shown but at least one length is correct, award M1 and deduct one A mark for each incorrect length. (no length correct and no working $\Rightarrow$ M0) If all 3 lengths are correct to at least 3 sf, award M1A1A1A0 If 2 are correct to at least 3 sf, award M1A1A0A0
(b)	
M1	Attempt to obtain any of the required angles. Method must be complete (ie reach a value for one angle) and formula used must be correct and values must be substituted into a correct formula.
A1	$\angle A = 90°$ Any labelling given can be ignored.
A1	$\angle B = \angle C = 45°$ All 3 correct w/o working scores M1A1A1
(c)	
M1	For indicating that the centre is at the midpoint of BC. This can be stated explicitly or used by attempting to find the midpoint. **OR:** Find equations for perpendicular bisectors of 2 of the sides and find the point of intersection
A1	Both coordinates correct. Correct answer written down w/o working scores M1A1
(d)	
M1	For indicating that the radius is half the length of BC. This can be stated explicitly or used by attempting to find half of their BC (not nec in the required form).
A1	Correct length of the radius, in the required form.
NB	If half the length of BC has been found earlier the marks for (d) can only be awarded if the length of the radius has been written in (d).

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Answer 1 of 8 · Question 3 part b · 4 marks

2019 JUN · Paper 02 · extract 6 of 11 · find part b in this mark scheme extract

Question Number	Scheme	Marks
9(a)	$f(x)=\frac{1}{4}x^4-\frac{1}{3}x^3-2x^2+4x\ (+c)$	M1A1
	$x=-2\ \ y=-\frac{28}{3}\ \ \Rightarrow c=0$	M1
	$\left(f(x)=\frac{1}{4}x^4-\frac{1}{3}x^3-2x^2+4x\right)$ $\therefore C$ passes through O	A1 cso (4)
(b)(i)	$x=2\ \ f'(x)=8-4-8+4=0$	M1
	$\frac{d^2y}{dx^2}=3x^2-2x-4$	M1
	$x=2\ \ \frac{d^2y}{dx^2}=12-4-4>0\ \ \therefore$ min at $x=2$	A1cso
	$x=1\ \ f'(x)=1-1-4+4=0$	M1
	$x=1\ \ \frac{d^2y}{dx^2}=3-2-4<0\ \ \therefore$ max at $x=1$	A1 cso
ALT	$f'(x)=(x-2)(x-1)(x+2)\ (=0)$ factorise	M1
	$x=2,1,(-2)$ solve (solutions to be 2,1 (and another))	M1
	OR: $f'(x)\ (=0)$ solved by calculator.	
	All 3 solutions needed (and correct) $=0$ not needed M2	
	$\frac{d^2y}{dx^2}=3x^2-2x-4$ differentiate	M1
	$x=2\ \ \frac{d^2y}{dx^2}=12-4-4>0\ \ \therefore$ min at $x=2$	A1cso
	$x=1\ \ \frac{d^2y}{dx^2}=3-2-4<0\ \ \therefore$ max at $x=1$	A1cso
(ii)	$x=1\Rightarrow y=1\frac{11}{12}\qquad x=2\Rightarrow y=1\frac{1}{3}$	B1B1 (7)
(c)	$y'=(x-1)(x-2)(x+2)$	M1
(i)	$x=-2,\ y=-\frac{28}{3}$ or $\left(-2,-\frac{28}{3}\right)$	A1
(ii)	$x=-2\ \ \frac{d^2y}{dx^2}=12+4-4>0\ \ \therefore$ min point	A1cso (3)
		[14]

Type MS · Distance travelled by integrating velocity

Answer 1 of 8 · Question 3 part b · 4 marks

2019 JUN · Paper 02 · extract 7 of 11 · find part b in this mark scheme extract

Question Number	Scheme	Marks
(a)M1	Attempt to integrate f'(x). The power of at least one x term must increase and none should decrease. c not needed	
A1	Correct integration, c not needed	
M1	Substitute the given coordinates to show $c = 0$. If c is not included (or assumed to be 0), then showing that substitution of $x = -2$ gives $y = -28/3$ is acceptable. Substitutions must be shown.	
A1cso	Correct conclusion from fully correct work. Accept eg $f(0) = 0$ $\therefore$ shown	
(b)	Ignore labels (i) and (ii) when marking (b)	
(i)M1	Substitute $x = 2$ in the expression for f'(x) to show f'(x) = 0. Substitution must be shown	
M1	Differentiate the expression for f'(x). At least one power must decrease and none increase.	
A1cso	Show second derivative is > 0 at $x = 2$ and give the conclusion. No errors or omissions in the working.	
M1	Substitute $x = 1$ in the expression for f'(x) to show f'(x) = 0. Substitution must be shown	
A1cso	Show second derivative is < 0 at $x = 1$ and give the conclusion. No errors or omissions in the working.	
(ii)B1	For either y coordinate correct (and x coordinate correctly indicated; substitution shown indicates this)	
B1	For the second y coordinate correct	
(c)	(May have been seen in (b))	
M1	Factorise f'(x) completely – any valid method OR use the factor theorem to find $x = -2$	
(i)A1	Extract the x coordinate of the third turning point and obtain the corresponding y coordinate. May quote y coordinate from the question	
(ii)A1cso	Test the sign of the second derivative at this point and make the conclusion. All work in (c) and $\frac{d^2y}{dx^2}$ (from (b)) must be completely correct for this mark to be awarded.	
	Alternative ways to determine the nature of the turning points:	
1.	If the change of sign of f'(x) is used then values of f'(x) either side of 1 and 2 must be calculated to provide evidence.	
2.	The continuity of a cubic function can be used to establish the nature of the turning points. If in doubt send to review.	

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Answer 1 of 8 · Question 3 part b · 4 marks

2019 JUN · Paper 02 · extract 8 of 11 · find part b in this mark scheme extract

Question Number	Scheme	Marks
10(a)	$\alpha+\beta=-3 \quad \alpha\beta=-5$	B1
(i)	$\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta, \ =19$	M1,A1
(ii)	$\alpha^4+\beta^4=(\alpha^2+\beta^2)^2-2\alpha^2\beta^2, \ =19^2-50=311$	M1,A1(5)
	OR: $\alpha^4+\beta^4=(\alpha+\beta)^4-4\alpha\beta(\alpha^2+\beta^2)-6\alpha^2\beta^2, \ =19^2-50=311$	
(b)	$(\alpha-\beta)^2=\alpha^2-2\alpha\beta+\beta^2=19+10$ OR $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta=9-(-20)$	M1
	$\alpha-\beta=\sqrt{29}$ *	A1 cso (2)
(c)	$\alpha^4-\beta^4=(\alpha^2-\beta^2)(\alpha^2+\beta^2)=(\alpha-\beta)(\alpha+\beta)(\alpha^2+\beta^2)$	M1A1A1 (3)
(d)	$\alpha^4-\beta^4=\sqrt{29}\times(-3)\times19=-57\sqrt{29} \ \left(-\sqrt{94221}\right)$	M1A1 (2)
(e)	$2\beta^4=\alpha^4+\beta^4-(\alpha^4-\beta^4)$	M1
	$\beta^4=\frac{1}{2}(311+57\sqrt{29}), =\frac{311}{2}+\frac{57}{2}\sqrt{29}$	A1,A1 (3)
	$p=\frac{311}{2} \quad q=\frac{57}{2}$	[15]
ALT	$\beta^4=\left(\frac{-3-\sqrt{29}}{2}\right)^4$ **and** use a correct binomial expansion M1A1	
	Correct final answer A1	

(a)B1	Correct sum **and** product of roots, seen explicitly or used (in (a)). Must be clear that sum is negative
(i)M1	Correct algebra, ready for substitution of sum and product
A1	Correct answer, condone use of $\alpha+\beta=3$.
(ii)M1	Correct algebra, ready for substitution
A1	Correct answer, condone use of $\alpha+\beta=3$.
(b)M1	Correct algebra and substitution of their values
A1cso	Correct answer from correct working. Must have seen sum = -3 here if not shown in (a)
(c)	
M1	Factorise to 2 quadratic brackets or 2 linear and one quadratic bracket
A1	Obtain 2 linear and 1 quadratic brackets with 2 of the 3 brackets correct
A1	Third correct bracket Accept $(\alpha^2+\beta^2)$ or $((\alpha+\beta)^2-2\alpha\beta)$
(d)M1	Substitute their values for each of the 3 brackets obtained in (c)
A1	Correct answer as shown or equivalent **exact** value
(e)	
M1	Correct expression for $2\beta^4$ or β^4
A1ft	Substitute their numbers to obtain a numerical expression for β^4 The expression must be exact but need not be simplified **NB** A correct numerical expression for their values implies M1
A1	Correct answer in the required form. p and q need not be shown explicitly.

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Answer 1 of 8 · Question 3 part b · 4 marks

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Question Number	Scheme	Marks
11(a)	$AC = \sqrt{(16^2+16^2)} = 16\sqrt{2}$	M1A1
ALT:	$AP^2 + PD^2 = 16^2 \Rightarrow AP = 8\sqrt{2}$	
	$VP = 8\sqrt{2}\tan 45 = 8\sqrt{2}$ (where P is the centre of the base) *	M1A1cso (4)
(b)	$VA^2 = \left(8\sqrt{2}\right)^2 + \left(8\sqrt{2}\right)^2 (= 256)$ or $VA = \dfrac{8\sqrt{2}}{\sin 45°}$	M1A1
	$VA = 16$ cm	A1 (3)
(c)	$DX^2 = 16^2 - 8^2$ where X is the foot of the perpendicular from D to VA	M1A1
	$DX = 8\sqrt{3}$	A1 (3)
(d)	$\tan\theta = \dfrac{8\sqrt{2}}{8}, \sin\theta = \dfrac{8\sqrt{2}}{8\sqrt{3}}, \cos\theta = \dfrac{8}{8\sqrt{3}}$	M1A1
	(or unsimplified if cosine or sine rule used) $\theta = 54.7°$	A1 (3)
(e)	$\cos\phi = \dfrac{\left(8\sqrt{3}\right)^2 + \left(8\sqrt{3}\right)^2 - \left(16\sqrt{2}\right)^2}{2\times 8\sqrt{3}\times 8\sqrt{3}} \left(= -\dfrac{1}{3}\right)$	M1A1
	$\phi = 109.5°$	A1 (3) [16]

(a)
M1 Use Pythagoras (with a + sign) to obtain the length of the diagonal of the base. Or use Pythagoras with correct sign to obtain the half diagonal
A1 Correct length for the diagonal or half diagonal
M1 Use tan in ΔAPV, their AP and angle of 45° to obtain the height
A1cso Correct answer with no errors in the working
OR: State ΔAVP is isosceles, or shown the 2 correct angles – can be on a diagram M1
$VP(=AP) = 8\sqrt{2}$ A1
OR use any other complete valid method M1 Correct result A1

(b)
M1 Use Pythagoras or trigonometry in ΔAPV or ΔAVC (or any other complete, valid method) to obtain a numerical expression for VA.
A1 Correct numbers in their choice of method.
A1 Correct length obtained.
$AV = 16$ w/o working scores M1A1A1

Type MS · Distance travelled by integrating velocity

Answer 1 of 8 · Question 3 part b · 4 marks

2019 JUN · Paper 02 · extract 10 of 11 · find part b in this mark scheme extract

Question Number	Scheme	Marks
(c)		
M1	Use Pythagoras with a minus sign (seen or implied) or trigonometry in ΔADX OR any other complete valid method **NB**: triangle ΔADV is equilateral.	
A1	Correct numbers in their choice of method.	
A1	Correct **exact** length for the perpendicular.	
(d)		
M1	Identify the correct triangle needed with the required angle marked (may be on Figure 1). This may be shown explicitly or implied by their work that follows.	
A1	Reach one of $\tan\theta = \frac{8\sqrt{2}}{8}$, $\sin\theta = \frac{8\sqrt{2}}{8\sqrt{3}}$, $\cos\theta = \frac{8}{8\sqrt{3}}$ oe	
A1	Correct answer, **must be 1 dp.**	
(e)		
M1	Use cosine rule: $\cos\phi = \frac{"DX"^2 + "XB"^2 - "BD"^2}{2\times"DX"\times"XB"}$ (their values)	
A1	Correct numbers substituted, follow through their previous answers	
A1	Correct answer, **must be 1 dp unless already penalised in (d)**	
	Any other routes should be marked: M1 Correct, complete method (ie it must be possible to reach a value for the required angle) A1 Correct numbers substituted A1 Correct answer, **must be 1 dp unless already penalised in (d)**	

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Answer 1 of 8 · Question 3 part b · 4 marks

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Type MS · Distance travelled by integrating velocity

Answer 2 of 8 · Question 1 part c · 3 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

1 (a)	$b^2-4ac=(-9)^2-4\times3\times7=-3$ $-3<0$ so no (real) solutions for v	M1 A1 [2]
(b)	$\left\{\frac{dv}{dt}\right\}=6t-9\Rightarrow a=6\times4-9=15\left(m/s^2\right)$	M1A1 [2]
(c)	$s=\int_0^5(3t^2-9t+7)\,dt=\left[t^3-\frac{9}{2}t^2+7t\right]_0^5$ $=5^3-\frac{9}{2}\times5^2+7\times5=\frac{95}{2}$(m) oe	M1 M1A1 [3]
		Total 7 marks

Part	Mark	Additional Guidance
(a)	M1	Applies the discriminant, or any other valid method. e.g. Attempts to solve the 3TQ by quadratic formula, Completing square, achieves $3\left(t-\frac{3}{2}\right)^2+...$ Attempts to solve $v = 0$, finds complex roots
	A1	Verifies that the equation for v has no (real) solutions By completing square, it must be correct, $3\left(t-\frac{3}{2}\right)^2+\frac{1}{4}$, states the graph is above x axis, or > 0 Finds correct complex roots $\frac{9\pm\sqrt{3}i}{6}$, states imaginary numbers/roots **Note**: Just stating "error" is not enough
(b)	M1	Attempts to differentiate the expression for v. Power of t to decrease in at least one term and increase in none.
	A1	Substitutes $t = 4$ and obtains correct acceleration, with or without unit.
(c)	M1	Attempt to integrate the expression for v. Power of t to increase in at least one term and decrease in none. Ignore limits if shown.
	M1	Clear substitution of the **given** limits into their **integrated** expression for distance, no need to see substitution of zero, can be implied by the correct distance found from a **correct integrated** expression
	A1	$\frac{95}{2}$ (m) oe e.g. 47.5, ignore unit

Type MS · Distance travelled by integrating velocity

Answer 2 of 8 · Question 1 part c · 3 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Distance travelled by integrating velocity

Answer 3 of 8 · Question 4 part c · 3 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

4 (a)	$v = \int (6t-12)\ \mathrm{d}t = 3t^2 - 12t\ (+c)$ When $t = 0$, $v = 0$ so $c = 0$ $v = 3t^2 - 12t = 3\times 2^2 - 12\times 2 = -12\ (\text{m/s})$	M1A1 A1cso [3]
(b)	$v = 0,\ 3t^2 - 12t = 0 \Rightarrow 3t(t-4) = 0 \Rightarrow T = 4$	M1A1 [2]
(c)	$s = \int (3t^2 - 12t)\ \mathrm{d}t = t^3 - 6t^2\ (+k)$ $d = \left\lvert \left[t^3 - 6t^2\right]_0^4 \right\rvert + \left[t^3 - 6t^2\right]_4^8 = 32 + 160 = 192\ (\text{m})$	M1 M1A1 [3]
Alternative		
(c)	$s = \int (3t^2 - 12t)\ \mathrm{d}t = t^3 - 6t^2\ (+k)$ At time $t = 0$, distance = 0, $k = 0$ $t = 4 \quad s = 4^3 - 6\times 4^2 = 64 - 96 = -32$ $t = 8 \quad s = 8^3 - 6\times 8^2 = 512 - 384 = 128$ Total distance travelled = 32 +32 +128 = 192 (m)	M1 M1 A1
		Total 8 marks

(a)	M1	Attempt to integrate $6t - 12$, must see power of t increased by one in either term and no decrease in powers. Do not accept $v = t(6t-12)$.
	A1	Correct result of integration. Constant is not required.
	A1cso	Correct velocity. Negative sign is required. Must show or state constant = 0.
(b)	M1	$3t^2 - 12t = 0$ and attempt to solve. ft their integration if M1 awarded in (a).
	A1	$T = 4$ Allow $t = 4$. Do not allow if $T = 0$ is also given as an option.
(c)	M1	Attempt to integrate their velocity providing it includes at least one term in t^2 or higher powers. Must see power of t increased by one in at least one term and no decrease in powers. Do not accept $s = t\times$ their v.
	M1	Attempt to use limits of 0 to 4 and 4 to 8 and then add the magnitudes of the two values. ft their $T = 4$ if M1 awarded in (b).
	A1	192
Alternative		
(c)	M1	Attempt to integrate their velocity providing it includes at least one term in t^2 or higher powers. Must see power of t increased by one in at least one term and no decrease in powers. Do not accept $v = t(6t-12)$.
	M1	Substitute and attempt to find distance when $t = 4$ and when $t = 8$. Working to confirm $k = 0$ must be shown. ft their $T = 4$ if M1 awarded in (b).
	A1	192

Type MS · Distance travelled by integrating velocity

Answer 3 of 8 · Question 4 part c · 3 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Distance travelled by integrating velocity

Answer 4 of 8 · Question 4 part b · 8 marks

2023 JUN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks
4(a)	$a=\frac{dv}{dt}=4t-16 \quad [t=5] \quad a=4\times5-16=4$	M1A1 [2]
(b)	$v=0\Rightarrow 2t^2-16t+30=0\Rightarrow (t-5)(2t-6)=0$ $\Rightarrow t_1=3,\ t_2=5$	M1 A1A1
	Displacement $=\int_3^5 (2t^2-16t+30)\,dt=\left[\frac{2t^3}{3}-\frac{16t^2}{2}+30t\right]_3^5$	M1A1
	$=\left[\frac{2\times5^3}{3}-8\times25+30\times5\right]-\left[\frac{2\times3^3}{3}-8\times9+30\times3\right]=\frac{100}{3}-36=-\frac{8}{3}$	M1A1
	Distance $=\frac{8}{3}$ (m)	A1ft
		[8]
		Total 10 marks

Type MS · Distance travelled by integrating velocity

Answer 4 of 8 · Question 4 part b · 8 marks

2023 JUN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	For an attempt to differentiate the given expression and substitute $t = 5$ See General Guidance for the definition of an attempt. Do not accept an expression with any terms integrated.
	A1	Obtains the value 4
(b)	M1	Sets $v = 0$ and attempts to solve the equation using a correct method. Please check their work carefully. See General Guidance for acceptable ways to solve a quadratic. If they use a calculator and only $t = 3, 5$ are seen with no working – award M1A1A1
	A1	For either $t_1 = 3$ or $t_2 = 5$
	A1	For both $t_1 = 3$ and $t_2 = 5$
	No working seen to find the distance.	
	SC: If a final answer of $\frac{8}{3}$ is seen with no evidence of algebraic integration, **award the final A mark only.**	
	M1	For an attempt to integrate the given expression for v with or without limits. Ignore poor or absent notation. [ie. No integral sign] See General Guidance for the definition of an attempt. Note: Do not accept an integrated expression with any **algebraic** terms differentiated.
	A1	For the correct integrated expression unsimplified or simplified. isw incorrect simplification following a correctly integrated expression. Accept for this mark the inclusion of $+ c$ Ignore limits for this mark.
	M1	Substitutes their values of $t_1 = 3$ or $t_2 = 5$ into their integrated expression and attempts to evaluate. Allow the subtraction either way around. Their integral must be a changed expression. Accept even if they have differentiated it. If the final answer is not $\pm\frac{8}{3}$ and there is no explicit substitution seen, award M0.
	A1	For the correct value of $-\frac{8}{3}$ You must see the negative value for this mark. Accept awrt -2.67 for this mark.
	A1ft	For the correct distance of $\frac{8}{3}$ (m) or exact equivalent. Do not accept 2.67 (m) but accept $2.\dot{6}$ **Note:** This is a ft mark. We will only award this mark if their value for the displacement is **negative** and the final distance is positive.

Type MS · Distance travelled by integrating velocity

Answer 4 of 8 · Question 4 part b · 8 marks

2023 JUN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Distance travelled by integrating velocity

Answer 5 of 8 · Question 4 part c · 4 marks

2025 NOV · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	B1	For $\frac{4x-3}{2x+1}$ or $\frac{2(2x+1)}{2x+1}-\frac{5}{2x+1}$ or $\frac{4x+2-5}{2x+1}$ **Both** correct asymptote equations seen will imply this mark.
(i)	B1	For $x=-\frac{1}{2}$ Independent of method accuracy mark, **so ignore any incorrect attempt to form a single fraction**
(ii)	B1	For $y=2$ Independent of method accuracy mark, **so ignore any incorrect attempt to form a single fraction**
(b)	B1ft	For the coordinates of the y-intercept, follow through their a or b and c from any expression of the form $\frac{4x+a}{2x+1}$ $a\neq 0$ or $\frac{4x+b+c}{2x+1}$ $b,c\neq 0$ Condone eg just $y=-3$ or $y=a$ without $x=0$ shown.
	B1ft	For the coordinates of the x-intercept, follow through their a as above. ie ft $(x=)-\frac{a}{4}$ Condone $x=\frac{3}{4}$ or $x=-\frac{a}{4}$ without $y=0$ shown.
Note on labelling: Answers presented in the correct order without labelling – we would award all marks. If labelling not present, we can't assume the candidate's thinking.		
(c)	B1	For a negative reciprocal curve drawn anywhere in the grid – there **must** be two branches present, they **must not** cross any asymptotes drawn or implied or cross each other and **must not obviously** 'bend back' on themselves. Mark intention.
	B1ft	Two clearly marked asymptotes, ft their asymptotes listed from part (a). Labelled as equations or clearly passing through correctly labelled numbers on the axes. Must be one section of negative reciprocal curve, tending towards these asymptotes. This branch must not obviously cross or bend back from any asymptotes drawn or implied. Mark intention. Ignore any other branch/curve/line drawn if there is one branch fulfilling these conditions.
	B1ft	Single curve (or even line) passing through the correct intercepts. Ft their coordinates from part (b) marked clearly on the graph as coordinates or crossing points on the axes. Ignore any other branch/curve/line present.
	The answers to (a) and (b) cannot be retrieved from (c)	

Type MS · Distance travelled by integrating velocity

Answer 5 of 8 · Question 4 part c · 4 marks

2025 NOV · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

Question	Scheme	Marks
4(a)	$\left(\frac{dv}{dt}=\right)8t-6=0\Rightarrow t=\left(\frac{6}{8}\right)$ oe	M1
	$v=4\times\left("\frac{6}{8}"\right)^2-6\times\left("\frac{6}{8}"\right)+5=\frac{11}{4}[\text{m/s}]$ oe	dM1A1 [3]
ALT	$4\left(t-\frac{3}{4}\right)^2+\frac{11}{4}$ oe	M1 M1
	$\frac{11}{4}[\text{m/s}]$ oe	A1 [3]
(b)	$18=8T-6\Rightarrow T=3$	M1A1 [2]
(c)	$\left[XY=\int_0^3\left(4t^2-6t+5\right)dt=\right]$	
	$\left\|\left[\frac{4t^3}{3}-\frac{6t^2}{2}+5t\right]_0^3\right\|$	M1A1
	$XY=\left(\frac{4\times3^3}{3}-\frac{6\times3^2}{2}+5\times3\right)-(0)=24\,(\text{m})$	M1A1 [4]
		Total 9 marks

Type MS · Distance travelled by integrating velocity

Answer 5 of 8 · Question 4 part c · 4 marks

2025 NOV · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract

Question	Scheme	Marks

Type MS · Distance travelled by integrating velocity

Answer 6 of 8 · Question 5 part c · 3 marks

2023 JUN · Paper 02 (regional R) · extract 1 of 5 · find part c in this mark scheme extract

5(a)	$\frac{dv}{dt} = 4t - 19$ $\frac{dv}{dt} = 4\times 5 - 19 = 1\left(\text{m/s}^2\right)$	M1 A1 [2]
(b)	$2t^2 - 19t + 35 = 0 \Rightarrow (2t-5)(t-7) = 0 \Rightarrow t = \frac{5}{2}, 7$ $t_1 = \frac{5}{2} \quad t_2 = 7$	M1 A1 [2]
(c)	$D = \int_{\frac{5}{2}}^{7} \left(2t^2 - 19t + 35\right) dt = \left[\frac{2t^3}{3} - \frac{19t^2}{2} + 35t\right]_{\frac{5}{2}}^{7}$ $\left(\frac{2\times 7^3}{3} - \frac{19\times 7^2}{2} + 35\times 7\right) - \left(\frac{2\times 2.5^3}{3} - \frac{19\times 2.5^2}{2} + 35\times 2.5\right)$ $= \left[\frac{49}{6} - \frac{925}{24}\right] = [8.167 - 38.542] = \left[-\frac{243}{8} \text{ or } -30.375\right]$ $\Rightarrow D = \frac{243}{8}$ (m) oe e.g. 30.375	M1 M1 A1 [3]
		Total 7 marks

Type MS · Distance travelled by integrating velocity

Answer 6 of 8 · Question 5 part c · 3 marks

2023 JUN · Paper 02 (regional R) · extract 2 of 5 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	M1	For differentiating the given v and substituting in $t = 5$ into the derivative. Both differentiation and substitution must be correct for this mark.
	A1	For the correct acceleration of 1 m/s^2. Units are not required.
(b)	M1	For attempting to solve the given 3TQ for the velocity. A correct method must be used, and they must reach **two** values of t for the award of this mark. See General Guidance for the definition of an attempt to solve a 3TQ. If there is no visible method seen, both values of t must be seen for the award of this mark.
	A1	For the two correct values of t They do not need to ne identified as t_1 or t_2 for this mark. Accept $\frac{5}{2}$ and 7 seen.
(c)	M1	For an attempt to integrate the given expression for v. At least one term must be correct and no terms are to be differentiated. If the value of 35 'disappears' it is M0. The question states 'use calculus' so integration must be seen.
	M1	For substituting the **two** values of t into their integrated expression. and subtracting the result (either way) of both substitutions. Allow for any changed expression from v A correct answer of $\pm\frac{243}{8}$ or ± 30.375 implies correct substitution into a correct integral. If the integration is incorrect, or the values are incorrect, full correct substitution must be seen for the award of this mark.
	A1	For the correct distance (which must be a positive value).

Type MS · Distance travelled by integrating velocity

Answer 6 of 8 · Question 5 part c · 3 marks

2023 JUN · Paper 02 (regional R) · extract 3 of 5 · find part c in this mark scheme extract

Question	Scheme	Marks

Type MS · Distance travelled by integrating velocity

Answer 6 of 8 · Question 5 part c · 3 marks

2023 JUN · Paper 02 (regional R) · extract 4 of 5 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	M1	For differentiating the given v and substituting in $t = 5$ into the derivative. Both differentiation and substitution must be correct for this mark.
	A1	For the correct acceleration of 1 m/s^2. Units are not required.
(b)	M1	For attempting to solve the given 3TQ for the velocity. A correct method must be used, and they must reach **two** values of t for the award of this mark. See General Guidance for the definition of an attempt to solve a 3TQ. If there is no visible method seen, both values of t must be seen for the award of this mark.
	A1	For the two correct values of t They do not need to ne identified as t_1 or t_2 for this mark. Accept $\frac{5}{2}$ and 7 seen.
(c)	M1	For an attempt to integrate the given expression for v. At least one term must be correct and no terms are to be differentiated. If the value of 35 'disappears' it is M0. The question states 'use calculus' so integration must be seen.
	M1	For substituting the **two** values of t into their integrated expression. and subtracting the result (either way) of both substitutions. Allow for any changed expression from v A correct answer of $\pm\frac{243}{8}$ or ± 30.375 implies correct substitution into a correct integral. If the integration is incorrect, or the values are incorrect, full correct substitution must be seen for the award of this mark.
	A1	For the correct distance (which must be a positive value).

Type MS · Distance travelled by integrating velocity

Answer 6 of 8 · Question 5 part c · 3 marks

2023 JUN · Paper 02 (regional R) · extract 5 of 5 · find part c in this mark scheme extract

Question	Scheme	Marks

Type MS · Distance travelled by integrating velocity

Answer 7 of 8 · Question 5 part c · 5 marks

2024 NOV · Paper 01 · find part c in this mark scheme extract

5 (a)	$3t^2-16t+5=0 \rightarrow (3t-1)(t-5) \rightarrow t =$	M1
	$t=\frac{1}{3}$ oe $\quad t=5$	M1 A1 [3]
(b)	$(a=)6t-16>0$	M1
	$t>\frac{8}{3}$ oe	A1 [2]
(c)	$\int_{(1)}^{(4)}(3t^2-16t+5)\,dt$	
	$\left[t^3-8t^2+5t\right]_{(1)}^{(4)}$	M1 A1
	$\left(4^3-8(4)^2+5(4)\right)-\left(1^3-8(1)^2+5(1)\right)$	M1
	-42	A1
	(distance) = 42	A1 [5]
		Total 10 marks

Type MS · Distance travelled by integrating velocity

Answer 8 of 8 · Question 3 part c · 4 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

ALT for first 3 marks: Uses the third term		
	B1	Let the 3rd term be T $20-T=d \quad T-8=d$
	M1	Solves the above equations simultaneously by any method to find d **AND** finds a value for a
	A1	For both correct $a=-4$ and $d=6$. Values may be implied by subsequent working
	M1	Correct substitution of their values for a and d into the correct formula for S_n (doesn't need to be placed > 220 at this point)
	M1	For the correct inequality shown with their values oe
	M1	For a full and correct method to solve their 3TQ leading to at least one value for n. Award this mark for the positive root [9.809…] without explicit working seen [use of calculator] following a correct 3TQ seen. If the 3TQ is incorrect and there is no evidence of explicit working to solve their 3TQ do not award this mark.
	A1	For choosing only positive root, and rounding to $n = 10$ If a candidate offers a positive and negative value without rejecting the negative [-7.47…] do not award this mark.
ALT: uses T & I Final two marks		
	M1	Attempts $n = 10$ **AND** $n = 9$ When $n = 10$ $S = 230$ and when $n = 9$ $S = 180$ **These values must be seen**
	A1	For choosing $n = 10$

Type MS · Distance travelled by integrating velocity

Answer 8 of 8 · Question 3 part c · 4 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks
3 (a)	$\left(v = 2t^2 - 19t + 35\right)$	
	$\left(\frac{dv}{dt} =\right) 4t - 19$	M1
	$4t - 19 = 0$	M1
	$t = \frac{19}{4}$	A1 [3]
(b)	$\left(v = 0 = 2t^2 - 19t + 35\right)$	
	$0 = (2t - 5)(t - 7)$	M1
	$T_1 = \frac{5}{2}$, $T_2 = 7$	A1, A1 [3]
(c)	$\left(\int\left(2t^2 - 19t + 35\right)dt\right) = \frac{2}{3}t^3 - \frac{19}{2}t^2 + 35t(+C)$ oe	M1 A1
	OR	
	$\int_{'2.5'}^{'7'}\left(2t^2 - 19t + 35\right)\,dt = \left[\frac{2}{3}t^3 - \frac{19}{2}t^2 + 35t\right]_{'2.5'}^{'7'}$	
	$\left(\frac{2}{3}(7)^3 - \frac{19}{2}(7)^2 + 35(7)\right) - \left(\frac{2}{3}\left(\frac{5}{2}\right)^3 - \frac{19}{2}\left(\frac{5}{2}\right)^2 + 35\left(\frac{5}{2}\right)\right) = -\frac{243}{8}$	M1
	$d = \frac{243}{8}$	A1 [4]
		Total 10 marks

Type MS · Distance travelled by integrating velocity

Answer 8 of 8 · Question 3 part c · 4 marks

2025 JUN · Paper 01 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks